MD. ROBAYET HASAN RUPOM

Computer Science & Engineering Student | Full-Stack Software Developer

Khulna, Bangladesh • +880 1714-058480 • rupomhossain58@gmail.com

GitHub: github.com/Rupom090 • Portfolio: rupom-portfolio.vercel.app

PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate with hands-on expertise in full-stack web application development and user-centered interface architecture. Proven record developing modular frontend applications with React 19, TypeScript, and Tailwind CSS, coupled with robust backend APIs using Laravel 12 and Python FastAPI. Experienced in designing relational database schemas, asynchronous queue processing systems, and responsive web workflows. Dedicated to engineering maintainable, accessible, and high-performance software solutions for complex real-world requirements.

TECHNICAL SKILLS

Programming Languages:	JavaScript (ES6+), TypeScript, PHP (8.2/8.3), Python 3, C, C++, HTML5, CSS3/SCSS
Frontend Development:	React (18/19), Vite, Inertia.js, Next.js, Tailwind CSS (v3/v4), shadcn/ui, Framer Motion, TanStack Query
Backend & API Engineering:	Laravel (11/12), FastAPI (Python), RESTful APIs, JWT Authentication, Laravel Sanctum, Laravel Horizon
Databases & DevOps Tools:	MySQL 8, Redis, Docker, Git, GitHub Actions, Postman, Cloudinary SDK
UI/UX & System Design:	Figma, Wireframing, User Flow Architecture, Interactive Mockups, Design Tokens, Usability Testing
Core Competencies:	Object-Oriented Design, Relational Data Modeling, Asynchronous Queue Pipelines, Web Accessibility (WCAG)

EDUCATION

Bachelor of Science in Computer Science & Engineering (B.Sc. in CSE) **Expected Graduation: 2027**

Northern University of Business and Technology Khulna (NUBTK) — Khulna, Bangladesh

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Systems, Software Engineering, Web Technologies, Computer Networks, Operating Systems, Human-Computer Interaction (HCI), Discrete Mathematics.

TECHNICAL PROJECTS

Humanova — AI Trust Governance & Verification Platform *React 19, TypeScript, Laravel 12, FastAPI, MySQL 8, Redis, Docker*

Repository: github.com/Rupom090/CSE4204-8C-T07-Humanova

- Architected an enterprise-grade AI trust verification platform featuring automated hallucination detection, URL/DOI citation validation, and an explainable multi-factor confidence scoring engine.
- Engineered a responsive dashboard in React 19 with Vite, TypeScript, and Tailwind CSS, integrating a Laravel 12 API gateway with an asynchronous Python FastAPI NLP microservice.
- Implemented background processing pipelines using Laravel Horizon and Redis, reducing claim extraction and semantic similarity scoring latency by **~35%**.
- Designed interactive multi-model comparison panels (OpenAI, Gemini, DeepSeek) with real-time benchmarking metrics and automated scan report exports.

TECHNICAL PROJECTS (CONTINUED)

AI Smart Study Planner for Students *React 18, TypeScript, Inertia.js, Laravel 11, PHP 8.2, MySQL, OpenAI*
Repository: **github.com/Rupom090/AI-Smart-Study-Planner-for-Students**

- Developed an end-to-end full-stack web application translating course syllabi and exam deadlines into structured daily study milestones and measurable focus task sessions.
- Integrated multi-modal AI tutoring tools (problem solver, essay grader, automated flashcards) with PDF parser text extraction, background queue workers, and JWT/Sanctum authentication.
- Engineered a seamless single-page application experience with React 18, Inertia.js, and Tailwind CSS featuring interactive progress analytics, Pomodoro timers, and session streaks.
- Designed resilient fallback mechanisms and retry policies, ensuring **99.9% uptime** and continuous study planning functionality during external AI API outages.

Research Hub UI — Academic Paper & Collaboration Portal *React, TypeScript, Tailwind CSS, REST APIs, Figma*
Repository: **github.com/Rupom090/research-hub-ui**

- Designed and developed an accessible UI/UX web portal tailored for academic researchers and students to search, bookmark, and collaboratively annotate scientific publications.
- Built **20+ reusable, WCAG-compliant UI components** in React and Tailwind CSS featuring faceted keyword filters, article abstract previews, and responsive layouts.
- Created comprehensive Figma design systems with reusable components, design tokens, and user flow prototypes prior to frontend implementation.

Wand-Enhancer — Advanced UX & Interoperability Tool *JavaScript (ES6+), TypeScript, DOM API, UI Optimization*
Repository: **github.com/Rupom090/Wand-Enhancer**

- Engineered a high-performance browser UX extension enhancing productivity shortcuts, workflow automation, and DOM interaction responsiveness.
- Optimized event listeners and client-side memory footprint, achieving **sub-16ms frame render times** and zero input latency during intensive DOM operations.

CERTIFICATIONS & PROFESSIONAL TRAINING

- **Full-Stack Web Development & Modern JavaScript:** Advanced full-stack application development, React component architecture, Node.js, RESTful API design, and modern ES6+ standards.
- **UI/UX Design Systems & Human-Centered Design:** Figma design token systems, component libraries, responsive grid architectures, and user testing methodologies.
- **Data Structures & Algorithmic Problem Solving:** Algorithmic complexity analysis, asymptotic optimization, recursion, dynamic programming, and competitive problem solving.
- **Backend Architecture & Relational Database Design:** Database normalization (1NF-3NF), query indexing, caching strategies with Redis, and secure token-based authentication.

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

- **University Technical Symposiums:** Actively presented technical projects, architectural overviews, and live software demonstrations at departmental programming sessions.
- **Open-Source Collaboration:** Maintained active GitHub repositories, adhering to semantic versioning, clean documentation, modular code structuring, and peer pull request reviews.
- **Continuous Skill Development:** Regular participation in competitive programming platforms and online developer communities to explore emerging web standards.

LANGUAGES

- **Bengali:** Native Proficiency
- **English:** Professional Working Proficiency (Technical communication, documentation, and academic presentation)